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Chapter 1: Artificial Neural Networks and Backpropagation

Artificial neural networks consist of interconnected layers of artificial neurons. During the forward pass, input features are multiplied by learnable weights, summed with a bias term, and transformed through non-linear activation functions such as ReLU.

Backpropagation applies the calculus chain rule to compute gradients of the loss function with respect to each parameter. Optimizers like AdamW update weights iteratively.

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Chapter 2: Transformer Architectures and Self-Attention

The Transformer model introduced by Vaswani et al. replaced recurrent networks with the Self-Attention mechanism. Scaled Dot-Product Attention allows models to weigh the importance of tokens across arbitrary sequence distances in parallel.

Multi-Head Attention projects queries, keys, and values into multiple subspaces, enabling modern models to capture nuanced grammatical and semantic relationships.